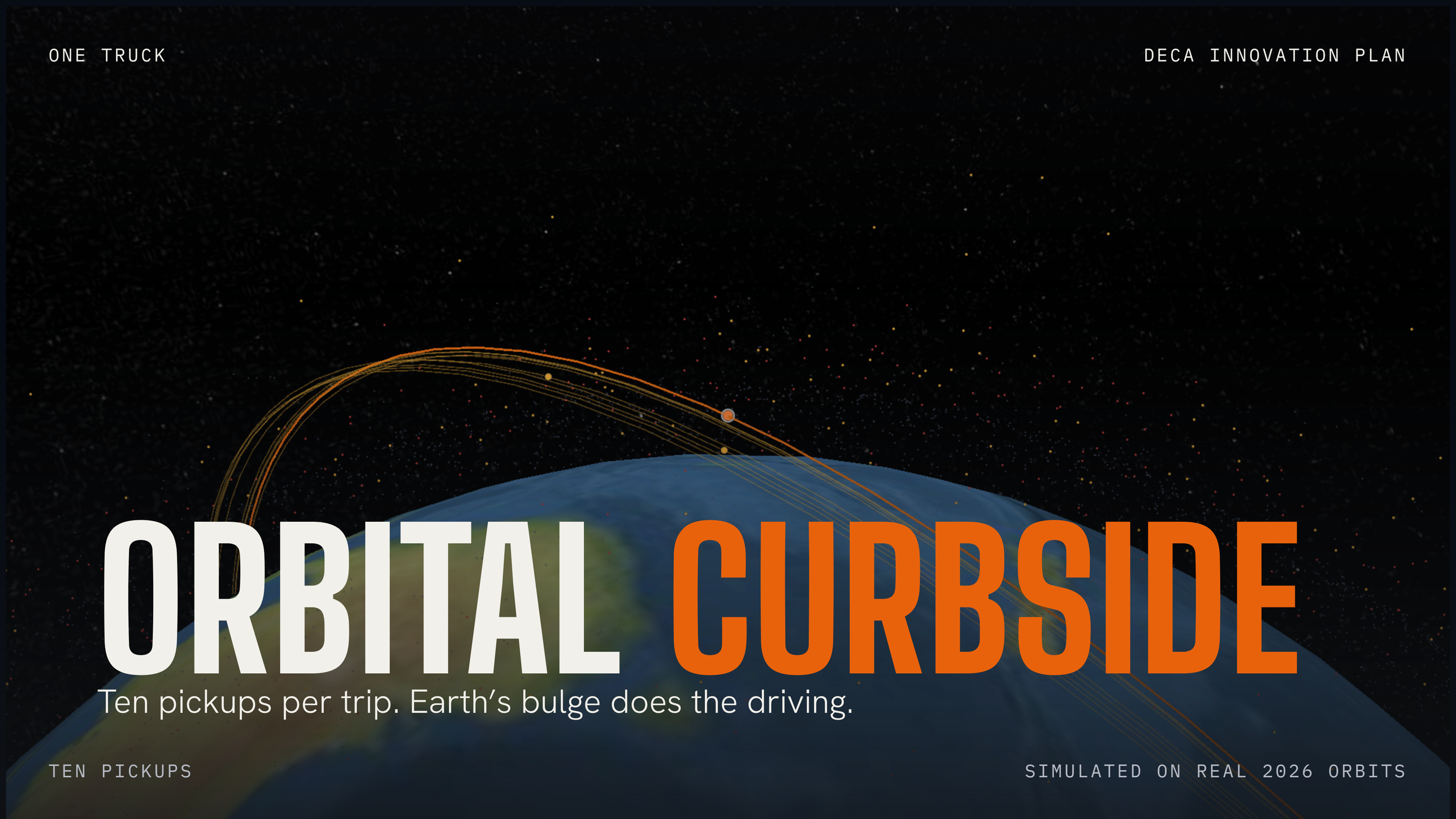


# ORBITAL CURBSIDE

Ten pickups per trip. Earth’s bulge does the driving.

INNOVATION PLAN

# Orbital Curbside

Removing dead satellites and rocket stages ten at a time, priced like a garbage route.




**Brandon Graff**
Founder & Chief Executive Officer

[School name]

[Street address]

[City, State ZIP]

[Presentation date]

DECA INNOVATION PLAN (EIP) · 2026–27

# Contents


I Overview 4

II Business Opportunity 5

III Customer Segments 8

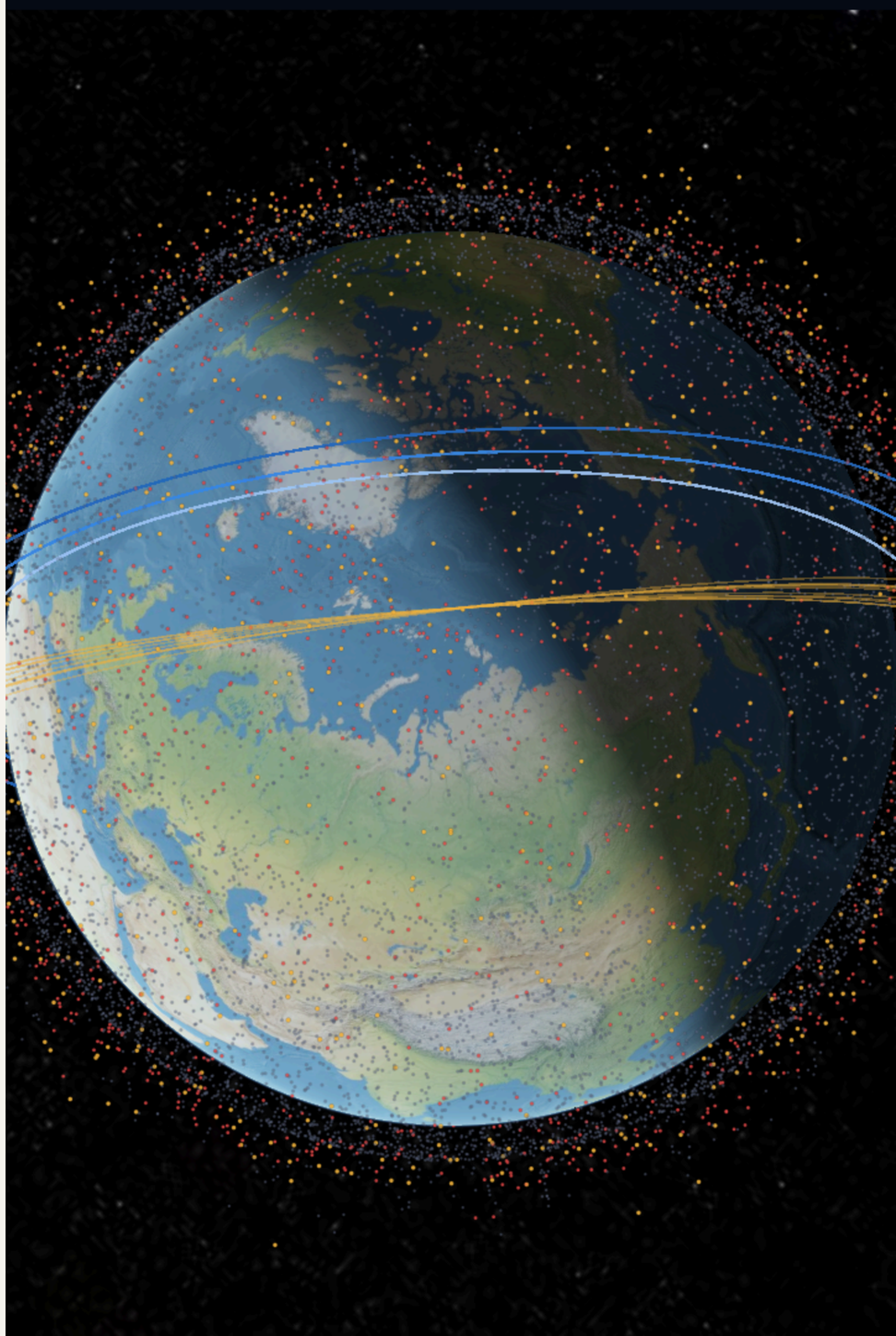
IV Unique Value Proposition 11

V Solution 13

VI Conclusion 18

VII Bibliography 19

VIII Appendix 20

# One electric spacecraft removes ten dead objects per trip, by letting Earth’s bulge do the driving.

## ABOUT

A space “garbage truck” that parks, drifts, hops and tags each object with a deorbit kit.

## BUSINESS MODEL

Removal as a service: a fixed fee per object, paid on confirmed deorbit.

## CUSTOMERS

**1** Space agencies
**2** Satellite operators
**3** Space insurers

## VALUE

$12.1M

modelled per removal, vs $40–93M per object in today’s contracts

# About 47,000 objects are tracked in orbit. ESA says the junk keeps multiplying even if we never launch again.



Our simulation, real catalogue of 16 Oct 2026 · ESA DISCOSweb (Oct 2026); ESA Space Environment Report 2026

# Today, removing one dead object is a custom mission costing $40–93 million.

Governments have started paying for removal, but they buy one or two objects at a time.

| Mission (buyer) | Objects | Public value |
|---|---|---|
| ClearSpace-1 (ESA, 2020) | 1 | €86M |
| ADRAS-J2 (JAXA, 2024) | 1 | ≈¥13.2B (≈$82M) |
| UK ADR (UK Space Agency, Oct 2026) | 2 | ≈£61M (≈$80M) |
| Disposal service (US SDA, Jan 2026) | 1 + options | $52.5M |

ESA; Astroscale; UK Government; Starfish Space press releases

# Why now: the rules just changed, and money is moving.

## 5 yrs

FCC limit to deorbit LEO satellites after their mission ends (for launches after Sept 2024).

## 0.9

Required disposal success (FCC Part 100, July 2026): up to 1 in 10 satellites may fail and still comply.

## $1.1–1.6B

Debris monitoring + removal market in 2025 (analyst estimates); removal alone ≈ $0.15B, growing ~39% a year.

FCC 22-74 (2022); FCC 26-47 (2026); Mordor Intelligence, ResearchAndMarkets, The Business Research Company

# Three customers, one route.

| Segment | What they need | How they buy | Their challenge |
|---|---|---|---|
| **1 · Space agencies**<br>ESA, JAXA, UKSA, NASA, US Space Force | Measurable collision-risk reduction per dollar; old rocket bodies gone | Competitive contracts, paid at milestones; want a demo first | Budget cycles; foreign-owned debris needs the owner’s consent |
| **2 · Satellite operators**<br>Iridium, Eutelsat OneWeb, Planet, Spire | A plan for satellites that die before they can deorbit; meet the 5-year rule | Per-object service; budget for end-of-life; insurance | Price-sensitive: today’s prices “don’t make sense” |
| **3 · Insurers**<br>space underwriters and brokers | A credible way to retire risk and pay claims | Removal bonds or assurance bundled with policies | Won’t commit until removal is demonstrated |

III · CUSTOMER SEGMENTS · OPERATORS (PRODUCT A)

# “I expect the cost is really in the millions or tens of millions, at which price I know it doesn’t make sense.”

Matt Desch, CEO of Iridium, on paying to remove its dead satellites (2019). He floated $10,000 per satellite.

## 24

dead first-generation Iridium satellites still in orbit in our Oct 2026 data (662 kg each)

## $12.1M

modelled cost per removal on a 10-stop Iridium route (555 days); a reusable truck pushes lower (≈$4.4M in our SL-8 model)

The gap to $10,000 is real. We close part of it now; we’re testing the rest in interviews.

Quote: C. Henry, SpaceNews, 30 Dec 2019 (via ABA Business Law Today, 2022) · CelesTrak SATCAT

III · CUSTOMER SEGMENTS · AGENCIES (PRODUCT B)

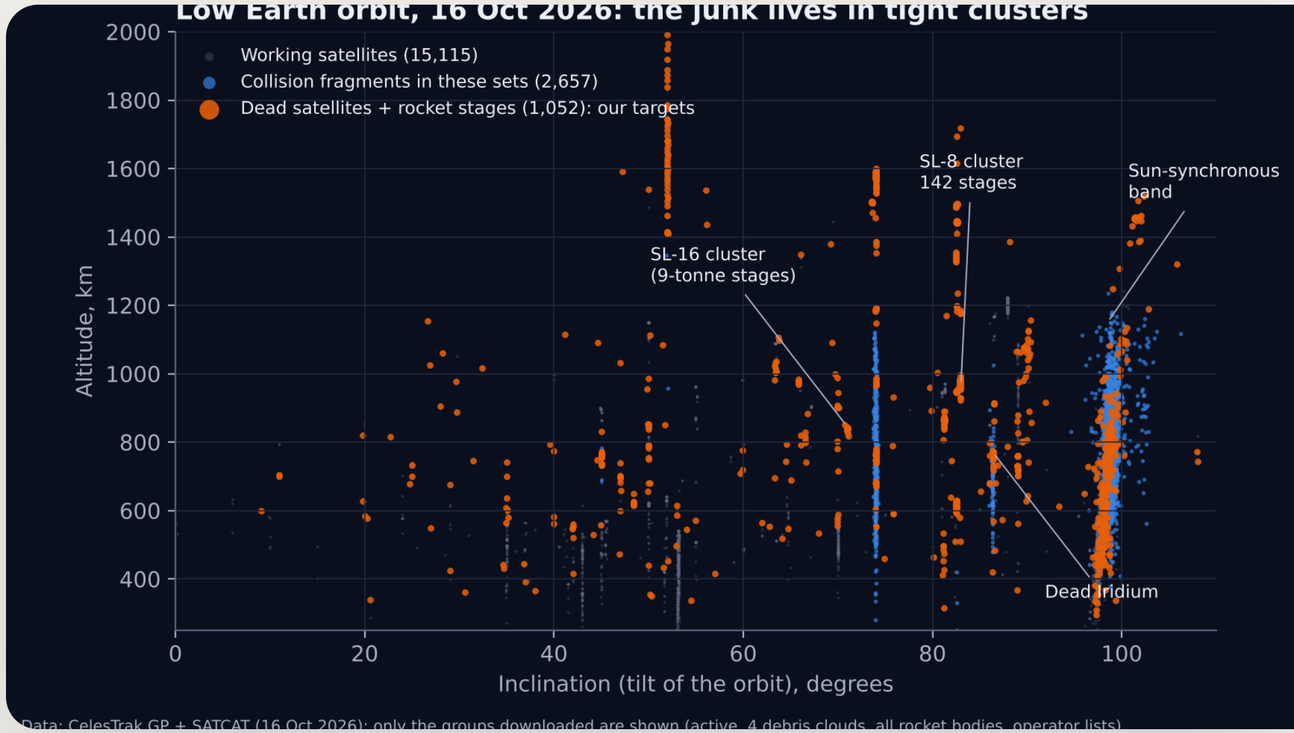
# 142 abandoned rocket stages share one orbit band. That’s a route.

The SL-8 cluster at 83°, ~960 km: about 204 tonnes of 1.4-tonne Soviet-era upper stages. SL-8 stages appear 11 times on the 2021 “50 most concerning” list. They won’t fall on their own for a century or more.

---

**What agencies need:** risk reduced per dollar, proof before scale, milestone payments.

**The catch:** these stages belong to Russia. Removal needs the owner’s consent, so we start with partner-nation objects and use this cluster as the technical model.




Our analysis of the CelesTrak catalogue, 16 Oct 2026; McKnight et al. (2021)

IV · UNIQUE VALUE PROPOSITION

# We don’t chase the junk. We let Earth bring it to us.

Space-debris removal priced like a garbage route: one truck, ten stops, a fixed fee per object, paid when it’s confirmed coming down.

IV · UNIQUE VALUE PROPOSITION

# A route beats a custom mission.

| | One-object missions (today) | Orbital Curbside |
|---|---|---|
| **Objects per launch** | 1–2 | **10 per route; ~100 per reusable truck** |
| **Cost per object** | $40–93M (public contracts) | **$12.1M modelled; ≈$4.4M with reuse** |
| **How it turns** | Burns fuel to change orbit planes | **Waits for Earth's bulge to line planes up** |
| **Engine** | Mostly chemical | **Electric (xenon): 56 kg for 10 stops** |
| **Payment** | Custom contract per mission | **Fixed fee per object, paid on confirmed deorbit** |
| **Planning** | Hand-designed per target | **Software picks the route from the live catalogue** |

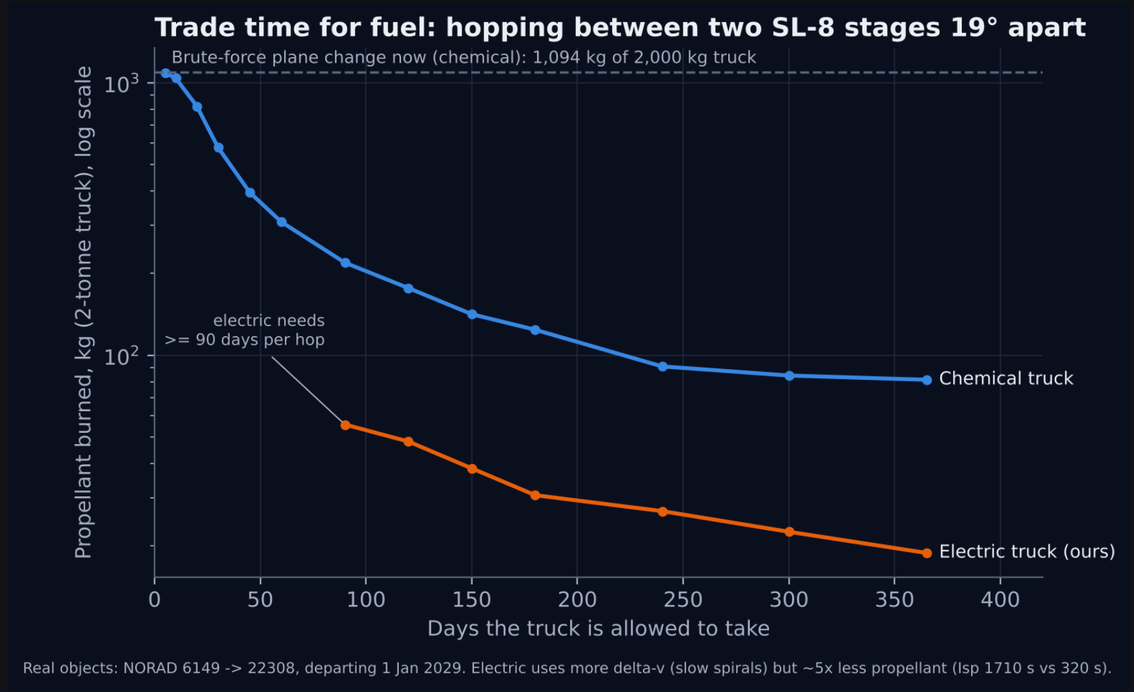
Contract values: slide 6. Our figures: model estimates, base assumptions (appendix).

01 **Park.** Move to a slightly different height than the next target.

02 **Drift.** Earth’s bulge turns the two orbits at different speeds, so the gap closes for free.

03 **Hop.** A short electric burn to meet the target.

04 **Tag.** Attach a small deorbit kit that drops the object into the atmosphere. Drive on.




Closing a 19° plane gap: burn now ≈ 1,094 kg of propellant; drift for a year ≈ 19 kg.

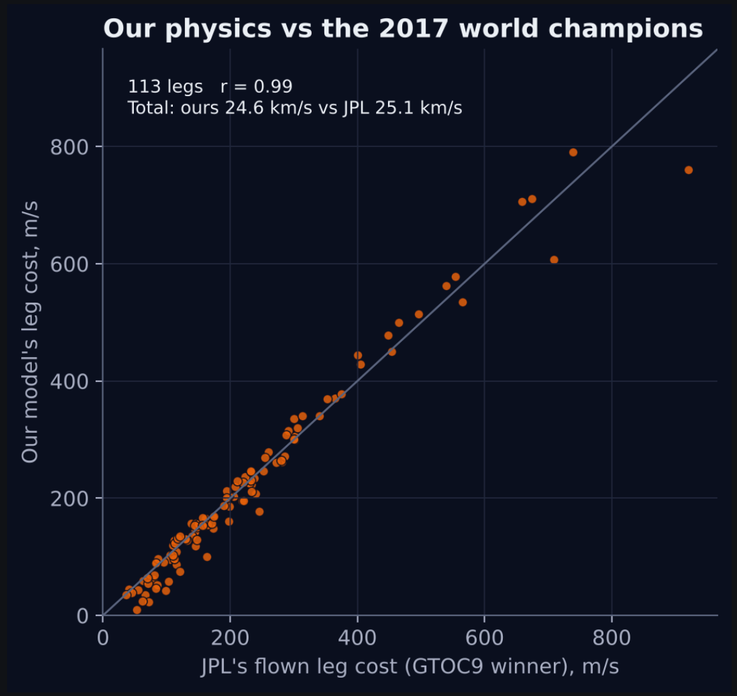
# We checked our math against NASA JPL’s winning solution.

**2.2%**
total Δv difference across JPL’s 113 legs (r = 0.99)

**0.06°**
median plane error vs SGP4, the model used to track real satellites

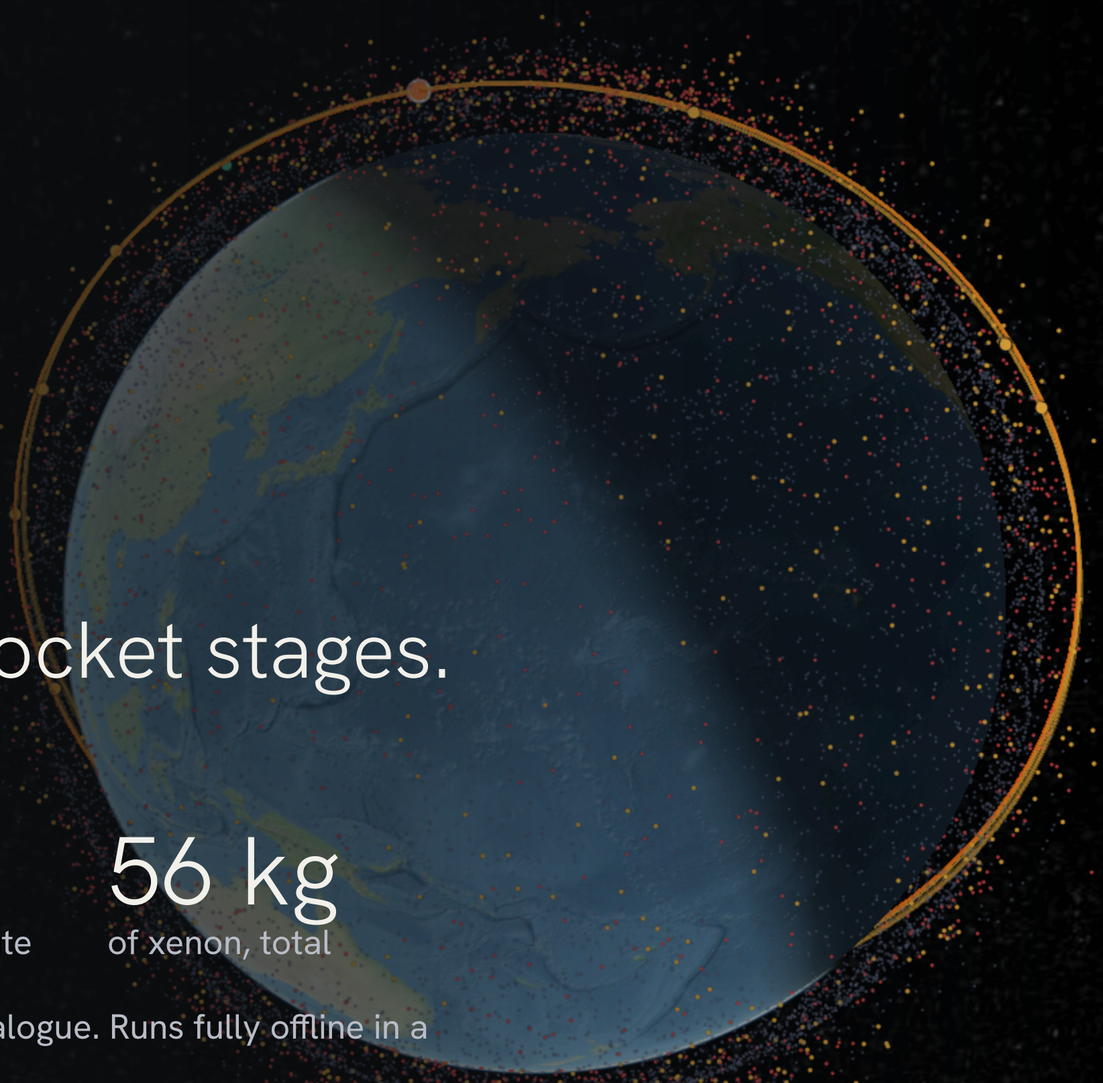
**2.6 m/s**
median correction per leg in a full-gravity check

**$2.4\times10^{21}$**
possible routes; four search methods compete to pick one




Each dot is one leg of JPL’s winning GTOC9 solution, recomputed by our model.

Petropoulos et al. (2018), GTOC9 JPL results; Vallado et al. (2006), SGP4



# One truck. Ten rocket stages. Seven months.

**10**
SL-8 stages removed

**220**
days on route

**56 kg**
of xenon, total

Real orbits from the 16 Oct 2026 catalogue. Runs fully offline in a browser.

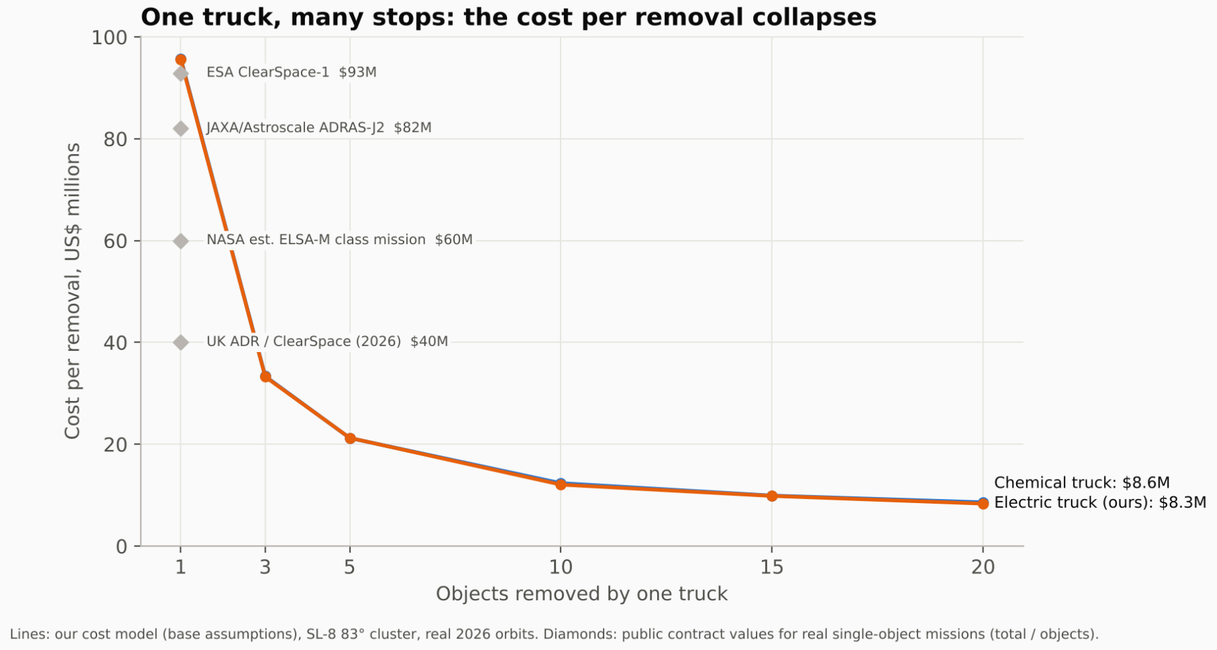
# More stops per trip, lower cost per object.

**$96M** one object per mission, same hardware

**$12.1M** 10-stop route (P10–P90: $11.0–15.2M)

**$4.4M** reusable truck, 10 routes, ~100 objects

Price at a 20% margin: $15.1M per object. Biggest unknown: truck hardware ($50–115M).




Cost per removal vs stops per route (our model)

Model estimates; inputs from NASA OTPS (2023), SpaceX rideshare pricing, Busek; assumptions in appendix

# What could stop us, and what we do about it.

### Capture

Grabbing a tumbling stage hasn't been done yet. We follow ADRAS-J2 and partner rather than invent the arm.

### Cost

Truck and kit prices are estimates. We test them with experts and keep a 10% fuel margin.

### Legal consent

Owners keep control of their objects. We start with customers' own and partner-nation objects.

### Re-entry safety

Heavy objects get a controlled re-entry over the ocean, not a slow fall.

**2026–27**

Validate: customer and expert interviews; refine the model

**2027–28**

Watch ADRAS-J2 capture; sign a capture and bus partner

**2028–29**

Pathfinder: 2–3 stops for one paying customer

**2029+**

First 10-stop truck

Outer Space Treaty (1967), Art. VIII · Astroscale ADRAS-J2

VI · CONCLUSION

# Removal priced like a route, not a moon shot.

1 Today’s removal costs $40–93M per object; our 10-stop route models $12.1M.

2 Earth’s bulge does the steering: 10 stops on 56 kg of xenon, checked against JPL and SGP4.

3 Impact: one reusable truck removes ~100 objects over ~8 years, about 13 a year.

THE ASK

[$__] in pre-seed funding for 12 months: finish customer validation, a kit design study, and a pathfinder partner.

Scan to fly the route

## VII · BIBLIOGRAPHY


ESA Space Debris Office. *Space Environment Statistics* (DISCOSweb), 14 Oct 2026; *Space Environment Report 2026*.

Kessler, D. J., & Cour-Palais, B. G. (1978). Collision frequency of artificial satellites. *J. Geophys. Res.*, 83(A6).

McKnight, D., et al. (2021). Identifying the 50 statistically-most-concerning derelict objects in LEO. *Acta Astronautica*, 181.

FCC. Second Report and Order, FCC 22-74 (2022); Report and Order, FCC 26-47 (22 Jul 2026).

ESA. ESA purchases world-first debris removal mission from start-up (1 Dec 2020).

Astroscale. Contract for Phase II of JAXA's CRD2 (20 Aug 2024).

UK Government. ClearSpace selected for UK Active Debris Removal mission (15 Oct 2026).

Starfish Space. Space Development Agency disposal contract (Jan 2026).

Colvin, T. J., Karcz, J., & Wusk, G. (2023). *Cost and Benefit Analysis of Orbital Debris Remediation*. NASA OTPS.

Henry, C. (2019, Dec 30). Iridium would pay to deorbit its 30 defunct satellites. *SpaceNews* (via ABA Business Law Today, 2022).

Mordor Intelligence; ResearchAndMarkets; The Business Research Company. Space debris removal market reports (2025).

United Nations (1967). Outer Space Treaty, Article VIII.

Izzo, D. (2017). Problem description for GTOC9: The Kessler Run. ESA. doi:10.5281/zenodo.570193

Petropoulos, A., et al. (2018). GTOC9: Results from the Jet Propulsion Laboratory. *Acta Futura*, 11.

Di Carlo, M., Romero Martin, J. M., & Vasile, M. (2017). Automatic trajectory planning for low-thrust active removal in LEO. *Adv. Space Res.*, 59(5).

Iskender, O. B., Felicetti, L., et al. (2026). Drift-aware multi-target space tug logistics. arXiv:2607.22974 (IAC-26).

Barea, A., Urrutxua, H., & Cadarso, L. (2020). Large-scale object selection and trajectory planning for multi-target ADR. *Acta Astronautica*.

Borelli, G., & Colombo, C. (2026). Space debris index definition for multi-purpose ADR. *Adv. Space Res.*, 77.

Kelso, T. S. CelesTrak GP data and SATCAT (retrieved 16 Oct 2026). celestrak.org

McDowell, J. C. General Catalog of Artificial Space Objects (15 Oct 2026). planet4589.org

Vallado, D. A., et al. (2006). Revisiting Spacetrack Report #3 (SGP4). AIAA 2006-6753.

Vallado, D. A. (2013). *Fundamentals of Astrodynamics and Applications* (4th ed.).

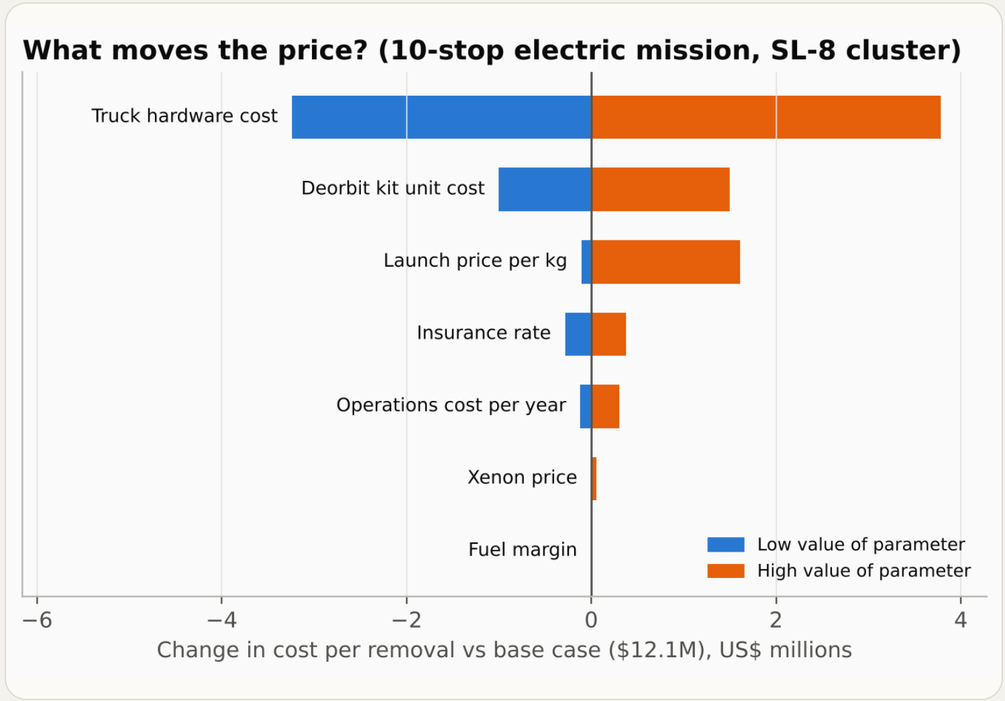
Edelbaum, T. N. (1961). Propulsion requirements for controllable satellites. *ARS Journal*, 31(8).

Busek Co. BHT-1500 Hall Effect Thruster datasheet (accessed Oct 2026).


Full list with links: written paper, references [1]–[24]

## VIII · APPENDIX · ASSUMPTIONS AND SENSITIVITY

| Input | Low · Base · High | Source |
|---|---|---|
| Truck hardware | $50 · 80 · 115M | NASA OTPS (2023); base = assumption |
| Deorbit kit, each | $0.5 · 1.5 · 3.0M | Assumption: no public price; testing in interviews |
| Launch | $7.0 · 7.5 · 15k/kg | SpaceX rideshare (2026–27); NASA OTPS |
| Xenon | $2.8 · 2.8 · 12k/kg | Market reports (2024–26) |
| Operations | $3 · 5 · 10M/yr | Assumption |
| Insurance | 5 · 8 · 12% | Assumption, of hardware + launch |
| Δv margin | 10 · 10 · 25% | Our JPL and SGP4 checks |
| Truck | 600 kg dry; Hall thrusters 0.202 N, Isp 1,710 s (Busek BHT-1500 class); 10 days per stop | |
| Kits | Solid motor, Isp 285 s; lower perigee to 250 km; ≈130 kg per SL-8 stage | |




Swing in cost per removal from each input, others at base. Truck hardware matters most ($8.8–15.8M).

Base case: SL-8 route, 10 stops, 220 days. Sources in paper.